



**INSTITUT KEMAHIRAN MARA LUMUT
UNIT PENILAIAN DAN PERSIJILAN**

**LO1 T2 THEORY ASSIGMENT
JANUARY-JUNE 2025 SESSION**

COURSE	: DFV 10023 – ELECTRONIC FUNDAMENTALS
DATE	: APRIL 2025
DURATION	: 1 WEEK
PROGRAMME	: DFD

Instructions :

1. This Assignment must be done by grouping.
2. Answer all the questions.
3. Meet all requirement.

This booklet consists 3 printed pages including this page.

INSTRUCTION

1. Writing an assignment report must be followed given criteria.

Format:

- a. Cover page
- b. Font: Calibri / Arial
- c. Size: 12
- d. Spacing: 1.15
- e. Header and footer: 0.4.”
- f. Margins: normal
- g. Paper: A4

The report must include:

- a) Reference

2. The assignment report does by group. Answers must include all procedures associated with each question given.

Submitting answer before or on **18 APRIL 2025**.

1. Explain Ohm's Law and its significance in electrical circuits.
2. What are the main differences between AC and DC current?
3. Describe the function and applications of a capacitor in electronic circuits.
4. What is a diode, and how does it function in a circuit?
5. Explain the working principle of a transistor and its role in electronic circuits.
6. What is the role of a resistor in an electronic circuit, and how is its value determined?
7. Describe Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL).
8. What is a Printed Circuit Board (PCB), and why is it important in electronics?
9. Explain the concept of a voltage divider and its practical applications.
10. What are the primary functions of an operational amplifier (op-amp) in electronic circuits?

END OF QUESTION